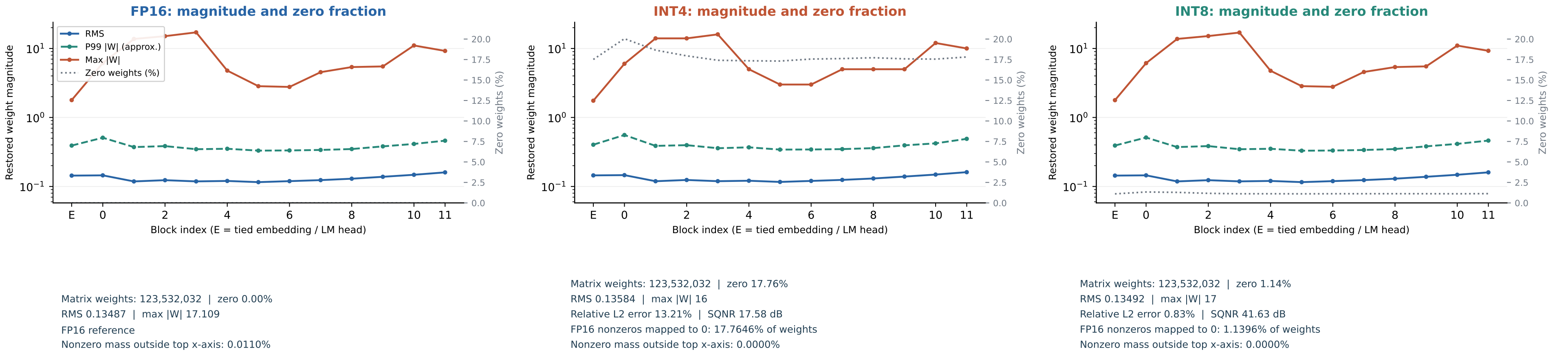
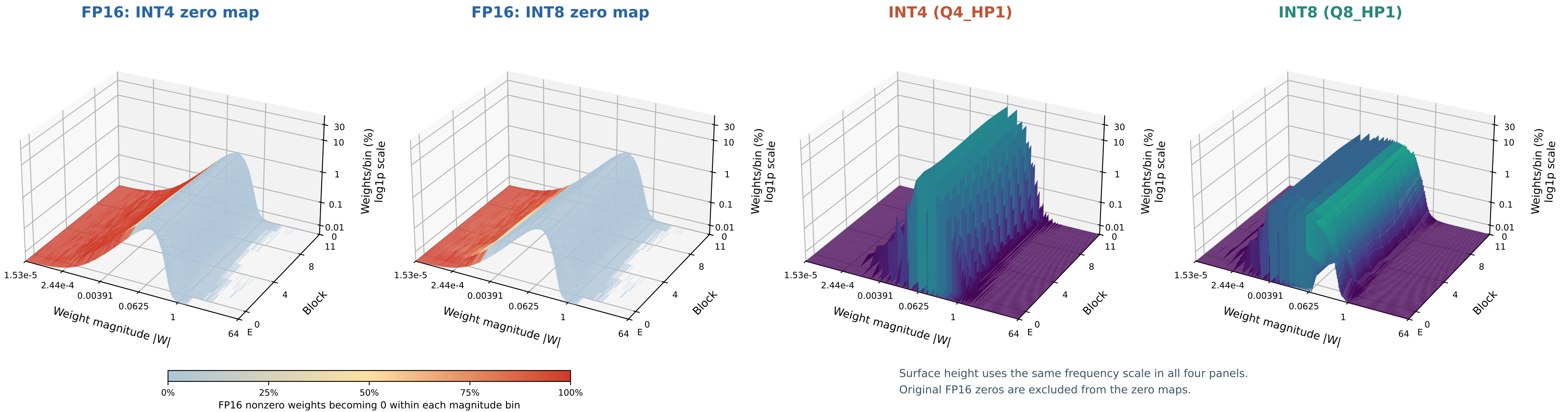


GPT-2 (124M) / FP16 vs PoTaI HP1

12 blocks | width 768 | FFN 3072 | MHA 12 heads | LayerNorm + GELU | learned positions

First two surfaces: original FP16 height, colored by the observed fraction mapped to zero. INT4 / INT8: restored weight distributions.

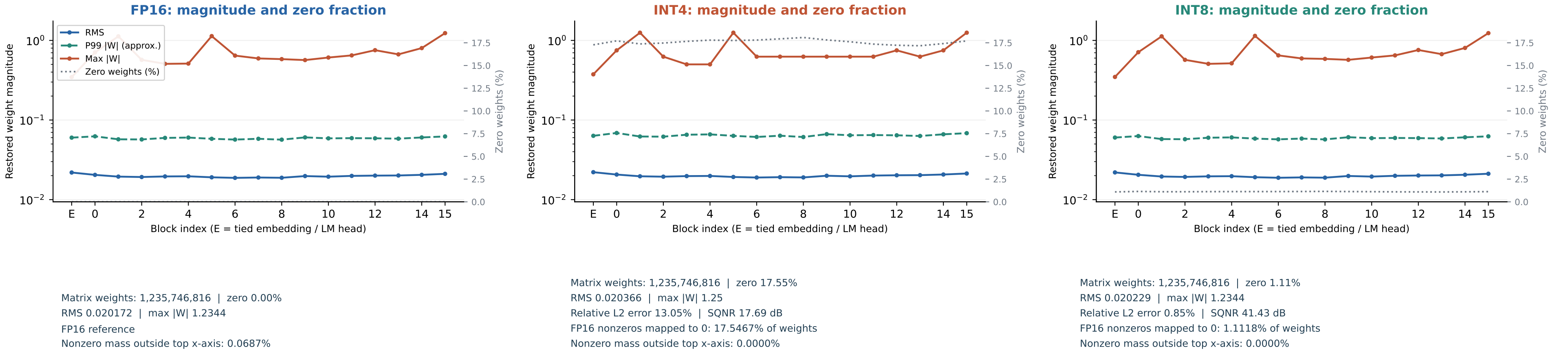
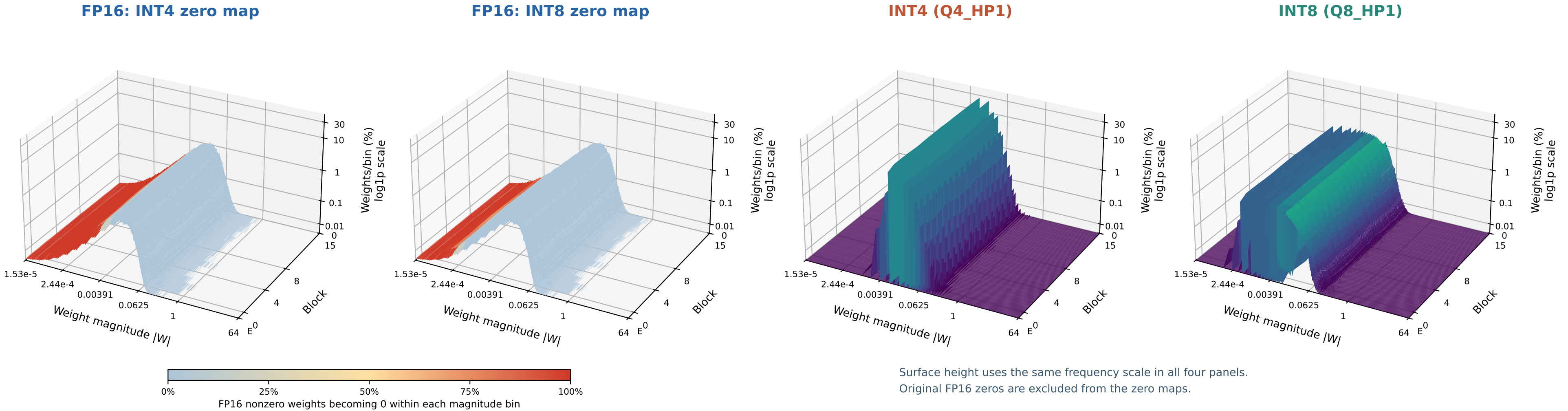


Zero maps compare the same weight positions: FP16 != 0 and decoded HP1 == 0, binned by the original FP16 magnitude. E = tied embedding / LM head.
Surface bin width: 1/32 log2. Frequency height = log10(1 + percent/0.01); ticks show actual percent. P99 uses coarse bins. No smoothing or bin downsampling.

Llama-3.2-1B / FP16 vs PoTal HP1

16 blocks | width 2048 | FFN 8192 | GQA 32 Q / 8 KV heads | RMSNorm + SwiGLU | RoPE

First two surfaces: original FP16 height, colored by the observed fraction mapped to zero. INT4 / INT8: restored weight distributions.



Zero maps compare the same weight positions: FP16 != 0 and decoded HP1 == 0, binned by the original FP16 magnitude. E = tied embedding / LM head.
Surface bin width: 1/32 log2. Frequency height = log10(1 + percent/0.01); ticks show actual percent. P99 uses coarse bins. No smoothing or bin downsampling.